

Vetting and installing trusted apps

Before any permissions management, start by installing apps only from vetted sources.

1. Use official app stores

- **Android:** Always download from Google Play to leverage Google Play Protect's daily malware scans and automatic threat detection.
- **iOS:** Install exclusively from the App Store, where Apple reviews every submission for malicious behavior and enforces notarization for enterprise apps.

2. Avoid sideloading whenever possible

- Sideloaded APKs bypass store vetting, elevating risk of trojans or backdoors.
- If you must sideload (eg, enterprise distribution), verify the APK's digital signature matches the developer's known key before installation.

3. Inspect app metadata and reviews

- On Android, tap About this app in Google Play to see developer info, last update date, and permission list.
- On iOS, review App Privacy details in the App Store listing to see exactly what data types the app collects and tracks.

Example: Before installing a new flashlight app, Raj checks that it requests only the CAMERA permission (to access the flash) and no access to contacts or location – an immediate red flag is present.



Managing runtime permissions

Controlling which apps can access sensitive sensors and data is central to minimizing risk.

Android permission controls

1. View and change app permissions

- Settings → Apps & notifications → See all apps → [App] → Permissions.
- Toggle each permission between Allow, Ask every time, or

Deny. For location, choose between Allow all the time, Allow only while in use, or Deny.

2. Auto-remove permissions for unused apps

- Android can automatically revoke permissions from apps you haven't used for months.
- Settings → Apps → [⋮] → Special app access → Unused apps → Turn on "Remove permissions if app isn't used".

3. Defer permission requests via intents

- If you're an app developer, use Android's Intent framework to defer reading contacts or files to system pickers, avoiding unnecessary manifest declarations.

Example: After installing a note-taking app, Priya immediately revokes its camera and microphone permissions, retaining only storage access for exporting notes.

iOS permission controls

1. Granting and revoking permissions

- Settings → Privacy & Security → [Category] (e.g., Location Services, Camera, Bluetooth).
- For each app listed, choose Never, Ask Next Time, or While Using the App.

2. On-demand permission requests

- Follow Apple's guideline to request permissions only when a feature is first used, not at install time – improving both privacy and user trust.

3. Monitor and reset permissions

- Settings → General → Transfer or Reset iPhone → Reset → Reset Location & Privacy to revert all permissions for a fresh start.

Example: When a social-media app first asks for camera access, Ananya taps Ask Next Time; only when she chooses to record a video is the permission granted.

Leveraging sandboxing and app containers

Sandboxing confines apps to isolated environments, preventing cross-app data leakage.

Android application sandbox

- Android assigns each app a unique Linux UID and private data directory, enforcing file-system and process isolation by default.
- Developer Step: Avoid over-granting sharedUserId in the manifest, which breaks isolation.

iOS app sandbox and entitlements

- iOS sandboxes apps with fine-grained entitlements declared